Memory Revival Sims & Layout

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1. Layout Screenshot

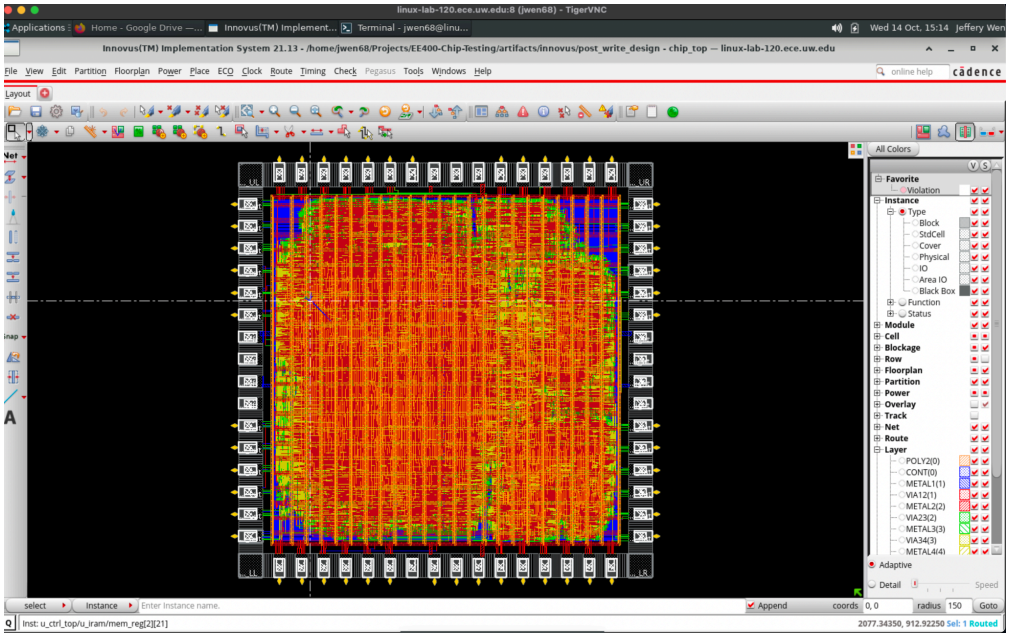


Figure 1: Layout

2. Simulation Screenshots

```
    [DBG 144820001000] uart_b=10 spi_reads=256 bsg_peer=185 boot_ok=1
    [DBG 144825001000] uart_b=10 spi_reads=256 bsg_peer=185 boot_ok=1
    [DBG 144830001000] uart_b=10 spi_reads=256 bsg_peer=185 boot_ok=1
    [DBG 144835001000] uart_b=10 spi_reads=256 bsg_peer=185 boot_ok=1
    [DBG 144840001000] uart_b=10 spi_reads=256 bsg_peer=185 boot_ok=1
    [DBG 144845001000] uart_b=10 spi_reads=256 bsg_peer=185 boot_ok=1
    [DBG 144850001000] uart_b=10 spi_reads=256 bsg_peer=185 boot_ok=1
    [DBG 144855001000] uart_b=10 spi_reads=256 bsg_peer=185 boot_ok=1

  ==== RESULTS ====
    firmware loaded           : /home/jwen68/Projects/EE400-Chip-Testing/chip/zeroscatter/firmw
  are.hex (164 words)
    SERV started (UART byte)  : 1
    i2c                       : (pad-observable via spec checker)
    ADC reads (SPI)           : 256
    UART bytes from chip      : 10
    BOOT OK seen              : 1
    BSG flits at TB peer      : 185
    last BSG flit at peer     : 0x0000
    0xB0 result packet seen   : 0
    PASS [firmware]: 'BOOT OK' seen on UART
    PASS [firmware]: 256 SPI ADC reads
    PASS [firmware]: 185 BSG flits reached peer

  ==== TEST_PASS ====
  $finish called from file "zeroscatter/chip_top_tb_syn.sv", line 613.

  ==== SPEC VERIFICATION RESULTS ====
    UART baud rate    : 47 bits correct, 0 violations
    UART 8N1 framing  : 10 frames correct, 0 violations
    I2C addr (0x5E)   :          PASS
    I2C data (0x7D)   :          PASS
    SPI CS duration   : 256 correct (16 SCK), 0 violations
    BSG flit format   : 1 bursts checked, 0 violations
    Mux sequencing    : 254 transitions correct, 0 violations
  ==== SPEC ALL PASS ====
  $finish at simulation time         144859930000
             V C S   S i m u l a t i o n   R e p o r t
  Time: 144859930000 ps
  CPU Time:     56.830 seconds;       Data structure size:   0.2Mb
  Thu Oct 15 14:34:21 2026
[jwen68@linux-lab-120 chip]$
```

Figure 2: Simulation Result.

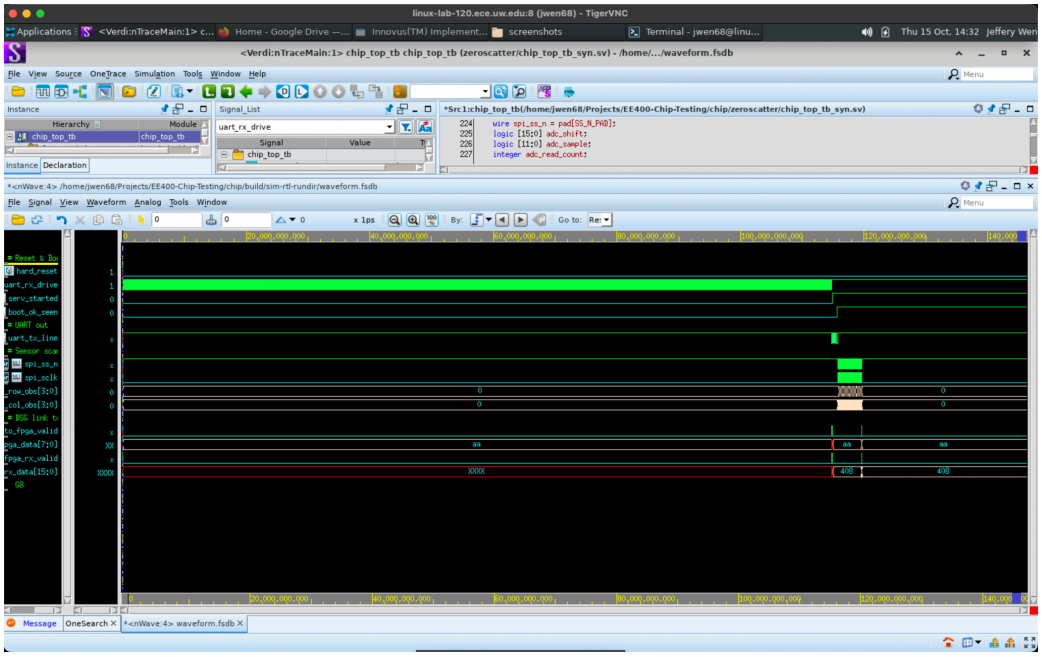


Figure 3: Simulation waveform.

3. Bring-up plan

We will test the chip step by step, starting with basic checks and then testing the full system.

Milestone 1: Any sign of life. Power up the chip (1.8 V core, 3.3 V IO), apply a clock and reset, and look for any activity on the outputs, such as the UART reply or the scan chain output. Any correct signal from the chip counts as success.

Milestone 2: Sub-modules work. Test each block on its own at a low clock (10 MHz):

- UART firmware loading and the "BOOT OK" message
- I²C sensor configuration
- SPI reads from the ADC and the mux scan across the 256 sensor cells
- the bsg_link chip-to-chip interface

Milestone 3: Test the full system at low speed. Run the complete process at 10 MHz: scan all 256 sensor cells, send the data to the conv chip, and receive the result through UART. If the results are correct, it shows that the full design works on the chip.

Milestone 4: Check the design targets. Increase the clock speed toward the original 50 MHz target and measure:

- The highest working clock speed at different supply voltages
- The frame rate (frames per second) of the full system
- Power use and energy used per frame

Milestone 5: Try other applications. After confirming that the chip works, explore its use in other sensing devices in Prof. Luo's lab, beyond the tactile glove.